

Chapter 2: The Sweep — Stage 1+2+3 Dynamics and the N=3..40 Runs (H $\perp$ /H Denominator-Control Figure)

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(Chapter 2 of the reproduction papers for the N=3..40 stage-1+2+3 sweep. Shares the Concept DOI 10.5281/zenodo.22317635 with the overview paper; Version DOI 10.5281/zenodo.22317636. Equation numbers continue from Chapter 1 (Eqs. 1–9). Code blocks are quoted verbatim from the programs; comments inside them remain in Japanese, as in the originals.)

1. Purpose

The central purpose of this series of numerical experiments is **to clarify the mechanism by which inflation-like development occurs — the phenomenon in which a seed too small to be measured (dormant fraction $H \perp / H \sim 10^{-30}$) is exponentially amplified by the dynamics alone over dozens of orders of magnitude until saturation**. Since this phenomenon was observed in the July canonical runs ($N = 40, 300, 1000$), the series has isolated, one factor at a time, which components of the dynamics are necessary conditions for this development (the confirmed result is the stage-1+2+3 composition; the audit record is Appendix A).

As the culmination, this chapter takes the 38 static parents ($N=3..40$) generated and verified in Chapter 1 as initial data, runs the stage-1+2+3 dynamics (phase-only, imaginary-part-only generator; real orthogonal rotation; fixed $\Delta \tau = 2 \pi / \text{den}$) for $N=3..40 \times 6$ denominators $\times 500$ steps, records at every step the indicator of inflation-like development, the **dormant fraction $H \perp / H$** (definition and its justification as an indicator: §2.5), the total energy H_{total} , and the zero closure $|z^{\wedge} T_z|/H$, and fixes how this development appears across the whole range of N and the sweep of the clock (denominator) in the target figure `fig_Hperp_denominator_controls_with_124_N3_N40_stage123.png`.

The descriptive purpose of this paper is complete reproducibility and the fixing of the correspondence among equations, programs, and data; it contains no physical interpretation beyond the observed facts. The grounds for the stage-1+2+3 composition are placed in **Appendix A (audit table of deletion controls)** and **Appendix B (program lineage)** so as not to interrupt the main text.

2. Theoretical Background

The dynamics of this chapter takes the dynamics of the July canonical (old) program as its starting point and reconstructs it on a new architecture, decomposed into three components: **stage 1**, **stage 2**, and **stage 3**. The theoretical background is therefore described in the order (2.1) the common state space,

(2.2) the mathematics of the old program's dynamics, (2.3) the mathematics of this program's dynamics (stage decomposition), (2.4) the mathematical differences between the old and new rotation maps, and (2.5) measured quantities and conservation laws.

2.1 State Space and Adjacency Structure (common to old and new)

(Eq. 10) Line-graph adjacency matrix — For edges $e=(i,j)$, $f=(k,l)$ of the complete graph K_N , $A_{ef} = 1$ when e and f share a vertex, 0 otherwise, 0 on the diagonal ($M \times M$, real symmetric). The edge ordering is the upper-triangular order identical to Eq. 1. The state is $z \in \mathbb{C}^M$ ($M = N(N-1)/2$); the edge phase is $\theta_e = \arg z_e$.

2.2 Mathematics of the Old Program (July Canonical) Dynamics

The old program is defined by the engine `run_n_scaling_lowrank_v1.py` bundled in this package (a bit-identical copy of the canonical). One step is the composition of the following three parts.

(Eq. 11) Old generator (phase-difference sine, real antisymmetric) — On vertex-sharing edge pairs (e,f) :

$$K_{ef}(\theta) = \cos \theta_e \sin \theta_f - \sin \theta_e \cos \theta_f = \sin(\theta_f - \theta_e)$$

The generator is rebuilt at every step from the current phases $\theta = \arg z$. The amplitudes $|z_e|$ do not enter. The vertex decomposition $K = C S^T - S C^T = W J W^T$ ($\text{rank} \leq 2N$) is the same as Eq. 3 of Chapter 1.

(Eq. 12) Power-iteration estimate of $\sigma_{\max}$ (approximate, history-dependent) — A real vector wp is carried across steps:

$$\begin{aligned} wp &\leftarrow -K(K wp) / \| -K(K wp) \| \quad (3 \text{ iterations}) \\ \hat{\sigma}_{\max} &= \| K wp \| \end{aligned}$$

Since $-K^2$ is positive semidefinite symmetric with largest eigenvalue $\sigma_{\max}^2$, $\hat{\sigma}_{\max}$ is a **power-iteration approximation** of $\sigma_{\max}$. Only 3 iterations are used, and wp is inherited from the previous step (warm start). Strictly speaking, the old dynamics is therefore not a map of z alone but a **system with a hidden state wp** , and $\hat{\sigma}_{\max}(t)$ is an approximate sequence depending on the trajectory history.

Engine implementation (`run_n_scaling_lowrank_v1.py`):

```

122     def sigma_max_power(self, wp, iters=3):
123         """warm-start 冪反復による  $\sigma_{\max}$  推定。wp は前ステップのベクトル (更新して返す)。"""
124         for _ in range(iters):
125             y = self.kmatvec(wp)
126             wp = -self.kmatvec(y)
127             nrm = np.linalg.norm(wp)
128             if nrm == 0.0:
129                 return 0.0, wp
130             wp = wp / nrm

```

```

131         sig = np.linalg.norm(self.kmatvec(wp))
132         return float(sig), wp

```

(Eq. 13) **Old one-step map (Cayley transform, σ -normalized clock)** — With the constant $\gamma = \tan(\pi/144)$:

$$K = K / \sigma_{\max}$$

$$z \leftarrow (I - \gamma K)^{-1} (I + \gamma K) z$$

Engine implementation (2N×2N solve via Woodbury):

```

134     def cayley_step(self, z, sigma):
135         """z ← (I-γ K)-1(I+ γ K) z, K = K/σ. Woodbury 0(N3). """
136         gn = GAMMA / sigma
137         r = z + gn * self.kmatvec(z)
138         A2 = (sigma / GAMMA) * self.J + self.G
139         rhs = self.wt(r)
140         y = np.linalg.solve(A2, rhs)
141         return r - self.w(y)

```

(Eq. 14) **Old eigenplane rotation angle** — The eigenvalues of K are purely imaginary pairs $\pm i \sigma_k$ ($\sigma_k \geq 0$). The Cayley factor for the eigenvalues $\pm i \sigma_k / \sigma_{\max}$ of K is

$$(1 + i \gamma \sigma_k / \sigma_{\max}) / (1 - i \gamma \sigma_k / \sigma_{\max}) \quad (\text{modulus } 1)$$

so the one-step rotation angle of eigenplane k is

$$\phi_k = 2 \arctan(\gamma \cdot \sigma_k / \sigma_{\max})$$

The rotation angle of the fastest plane ($\sigma_k = \sigma_{\max}$) is always fixed at $\phi_{\max} = 2 \arctan(\tan(\pi/144)) = \pi/72$. This is an **adaptive (σ -normalized) clock** that “ticks time by the system’s fastest mode”, and the spectral dependence of the rotation angle undergoes **nonlinear compression** by $\arctan$.

(Eq. 15) **Old conserved quantities** — Since K is real antisymmetric, the Cayley transform $O = (I - \gamma K)^{-1}(I + \gamma K)$ is a **real orthogonal matrix** ($O^T O = I$). A real orthogonal matrix acts identically on the real and imaginary parts of $z = x + iy$, so

$$\|Oz\|^2 = \|z\|^2 \quad (\text{norm conservation})$$

$$(Oz)^T(Oz) = z^T O^T O z = z^T z \quad (\text{exact conservation of the zero closure } z^T z)$$

(Eq. 15’) **Relative equilibrium (parent)** — The parent v lies on an eigenplane of K ($\arg v$) (Chapter 1, Eqs. 4–6); on the flow $dZ/dt = KZ$ we have $KZ = -i \sigma Z$, i.e., $Z(t) = e^{\{-i \sigma t\}} Z(0)$. All edge phases advance by the same amount, so phase differences are invariant, hence $K(\theta(t)) = K(\theta(0))$. The parent is a relative equilibrium that “rotates rigidly without changing phase differences”.

2.3 Mathematics of This Program’s Dynamics (Definitions of Stages 1, 2, 3)

The stage numbering follows the definitions of the $N=40$ single-factor experiment series (Appendix A).

Stage 1 provides the baseline architecture, and stages 2 and 3 are the two modifications applied to it.

(Eq. 16) **Stage 1: baseline architecture (explicit matrix, spectral map, fixed $\Delta \tau$ clock)**

— The generator is constructed explicitly as a Hermitian matrix H , and one step is given by the exact spectral exponential map

$$H = V \text{diag}(w) V^\dagger, \quad z \leftarrow V e^{-i \Delta \tau w} V^\dagger z$$

The clock is a **fixed external clock**

$$\Delta \tau = 2 \pi / \text{den}, \quad \text{den} \in \{N-2, N-1, N, N+1, N+2\} \cap \mathbb{N}^+ \cup \{124\}$$

(For small N the series becomes special due to the condition $\text{den} > 0$. For $N=3$ the series is $\text{den}=1,2,3,4,5,124$; $\text{den}=1$ means $\Delta \tau = 2 \pi$.) In the stage-1-only baseline, $H = A^*(\bar{z} \otimes z)$ (amplitude-weighted) is used. This map is memoryless (one step is a pure function of z and constants); the old hidden state w_p and approximate σ estimation do not exist.

■2.3.1 Design Background (Hypothesis) of the Denominator Series and Its Implementation Correspondence

The denominator series $\{N-2, N-1, N, N+1, N+2, 124\}$ of $\Delta \tau = 2 \pi / \text{den}$ is not an invention of this chapter; it inherits, unchanged, the design of the **denominator control experiments** of this series (2026-09-03; design instruction document `ChatGPT_denominator_controls_N3_N40_mixedseed_20260903/CLAUDE_CODE_RUN_I`). The background splits into two lines.

(a) **den=124 (fixed-clock control)** — Inherited from the step-size convention $\Delta \tau = 2 \pi / L$, $L=124$ of the interference-preserving dynamics series (e.g., line 27 $L=124$ of `.../program/pass1_parents.py` in the qualification-and-seedless series folder). It is the control with the clock fixed independently of N (legacy convention).

(b) **den $\in \{N-2..N+2\}$ (hypothesis series of a system-scaled clock)** — The hypothesis connects to the implementation through the rotation angle of Eq. 20, $\psi_k = (2 \pi / \text{den}) \cdot \sigma_k$. Measuring $\sigma_{\max}$ from the parents' σ spectra saved in Chapter 1 (the `sigma` field of the `npz`; aggregated in `analysis_sweep_summary_v1.json`, `sigma_max_by_N`):

N	$\sigma_{\max}$	$\psi_{\max}/2 \pi$ (den= N)	$\psi_{\max}/2 \pi$ (den=124)
3	1.414	0.471	0.011
5	3.742	0.748	0.030
10	8.928	0.893	0.072
20	18.894	0.945	0.152
30	28.898	0.963	0.233
40	38.905	0.973	0.314

$\sigma_{\max}(N)$ grows nearly linearly with N ($N - \sigma_{\max} \approx 1.1$). Therefore:

- In the **den $\approx N$ series**, the one-step rotation of the fastest eigenplane is placed at $\psi_{\max} \approx 2 \pi \cdot (1 - O(1)/N)$, i.e., in the **nearly-one-full-turn (near-commensurate with 2π , stroboscopic) regime**. The five points $\text{den} = N, N \pm 1, N \pm 2$ sweep the detuning from this commensurability.

- With **den = 124**, all eigenplanes stay in the small-angle regime ($\psi_{\text{max}}/2 \pi \leq 0.314$ within this sweep), a control close to the small-step approximation of the continuous flow $dz/dt = Kz$ (**flow-like regime**).

Implementation correspondence: den is determined by the series generation on line 38 (`pairs=[(N+o,...) for o in OFFSETS if N+o>0]+[(124,'124')]`) and enters the dynamics only through $2.0 \cdot \text{math.pi} / \text{den}$ on line 27. den is used nowhere else (as Eq. 20 states, the effect of den is exhausted by the linear scaling of rotation angles and aliasing).

This chapter does not judge this design hypothesis. The den-dependent observations are recorded as facts in §6 (denominator dependence of onset; distribution of non-crossing runs).

(Eq. 17) Stage 2: amplitude normalization — The input to the generator construction is replaced by the unit-modulus $\hat{z}$:

$$\begin{aligned}\hat{z}_{\text{e}} &= e^{i \theta_{\text{e}}} = e^{i \arg z_{\text{e}}} \\ \hat{H}_{\text{ef}} &= A_{\text{ef}} \cdot \text{conj}(\hat{z}_{\text{e}}) \quad \hat{z}_{\text{f}} = A_{\text{ef}} \cdot e^{i(\theta_{\text{f}} - \theta_{\text{e}})}\end{aligned}$$

The generator thereby becomes a function of the **phases only**, as in the old dynamics (Eq. 11). The amplitudes of the state z itself are retained and continue to evolve as dynamical variables (the normalization happens only inside the generator construction, at every step).

(Eq. 18) Real/imaginary decomposition of $\hat{H}$ — $\hat{H}$ is Hermitian and decomposes uniquely into a real symmetric part and a real antisymmetric part:

$$\begin{aligned}\hat{H} &= S + iK, \\ S_{\text{ef}} &= A_{\text{ef}} \cos(\theta_{\text{f}} - \theta_{\text{e}}) \quad (\text{real symmetric}) \\ K_{\text{ef}} &= A_{\text{ef}} \sin(\theta_{\text{f}} - \theta_{\text{e}}) \quad (\text{real antisymmetric})\end{aligned}$$

This K is the same matrix as the old generator (Eq. 11).

(Eq. 19) Stage 3: extraction of the imaginary part (removal of the cos symmetric part) and real orthogonal rotation — Instead of $\hat{H}$, the generator

$$H_3 = i K = i \cdot \text{Im}(\hat{H})$$

is used ($i \times \text{real antisymmetric} = \text{Hermitian}$). Substituting into the stage-1 map (Eq. 16):

$$z \leftarrow \exp(-i \Delta \tau \cdot iK) z = \exp(\Delta \tau K) z$$

$\exp(\Delta \tau K)$ is the exponential of a real antisymmetric matrix, i.e., a **real orthogonal matrix**. It therefore has the same conserved quantities $\|z\|^2, z^T z$ as the old dynamics (Eq. 15; same proof). Since stage 2 makes K phase-only, the relative-equilibrium argument of Eq. 15' also holds unchanged: along the parent orbit K is time-invariant and the parent rotates rigidly.

(Eq. 20) New eigenplane rotation angle — The eigenvalues w_k of $H_3 = iK$ are real; corresponding to $\pm i \sigma_k$ of K , $w = \pm \sigma_k$. The one-step rotation angle of eigenplane k is

$$\psi_k = \Delta \tau \cdot \sigma_k = (2 \pi / \text{den}) \cdot \sigma_k$$

The rotation angle is **linear** in the eigenvalue, with no upper bound. The phase factor $e^{-i \Delta \tau w}$ wraps around in w with period $2 \pi / \Delta \tau$ (**aliasing**): for eigenplanes with $\Delta \tau \sigma_k$ exceeding 2π , the

effective rotation angle is $\Delta \tau \sigma_k \bmod 2\pi$.

2.4 Mathematical Differences between the Old and New Rotation Maps (the crux of this chapter’s dynamics)

Both are “real orthogonal rotations generated by the phase-only real antisymmetric K ”, but **the spectral map of rotation angles and the clock differ mathematically**.

Item	Old (Eqs. 12–14)	This chapter = stages 1+2+3 (Eqs. 16–20)
Eigenplane rotation angle	$\phi_k = 2 \arctan(\gamma \sigma_k / \sigma_{\max})$	$\psi_k = (2\pi/\text{den}) \sigma_k$
Spectral dependence	nonlinear compression by $\arctan$ ($ \phi < \pi$)	linear (with mod-2 π aliasing)
Clock	$\sigma_{\max}$ -normalized (fastest plane fixed at $\pi/72$; system-intrinsic)	fixed $\Delta \tau = 2\pi/\text{den}$ (external; den is the sweep parameter)
Obtaining σ	approximate $\sigma_{\max}$ by 3 power iterations (warm start, history-dependent)	exact eigenvalues via eigh (no estimation)
Realization of the map	Cayley rational form applied via Woodbury solve ($O(N^3)/\text{step}$)	exact exponential applied via spectral decomposition ($O(M^3)/\text{step}$)
Memory	carries hidden state wp (not a pure function)	memoryless (pure function of z)
Conserved quantities	$\ z\ ^2, z^T T z$ (real orthogonal)	same (real orthogonal)
Relative equilibrium (parent)	holds (Eq. 15')	holds (consequence of Eq. 19)

(Eq. 21) Correspondence in the small-angle limit — In the regime $\Delta \tau \sigma_k \ll 1$ and $\gamma \sigma_k / \sigma_{\max} \ll 1$, $\arctan x \approx x$ gives

$$\phi_k \approx (2\gamma/\sigma_{\max}) \cdot \sigma_k = \Delta \tau_{\text{eff}} \cdot \sigma_k, \quad \Delta \tau_{\text{eff}} = 2 \tan(\pi/144) / \sigma_{\max} \approx (\pi/72) / \sigma_{\max}$$

That is, the old dynamics approximately coincides with “Eq. 20 with $\Delta \tau_{\text{eff}}$ re-normalized by $\sigma_{\max}$ at every step”. The differences between the two reduce to (i) the value of $\Delta \tau$ (externally fixed $2\pi/\text{den}$ vs. the system-intrinsic $(\pi/72)/\sigma_{\max}$), (ii) the presence of $\arctan$ compression and aliasing, and (iii) the exactness of σ . The measured record of this correspondence is the last row of Appendix A (the σ -normalized-clock reference experiment: late-segment slope 65.8 steps/decade vs. the July canonical 64.0 steps/decade). The runs of this chapter use the **fixed $\Delta \tau$ (stage 1)** throughout; the sweep of den measures the effect of this clock difference.

2.5 Measured Quantities and Conservation Laws — Definition of the Dormant Fraction

$H_{\perp}/H$ and Its Justification as the Inflation Indicator

(Eq. 22) **Readout plane** — From the initial state $z_0 = v + \delta g$ (after normalization):

$$p = \text{Re } z_0 / \| \text{Re } z_0 \|, \quad q' = \text{Im } z_0 - (\text{Im } z_0 \cdot p) p, \quad q = q' / \| q' \|$$

p, q are **fixed** throughout the run (the Gram–Schmidt orthonormal basis of the real 2-dimensional plane $\Pi = \text{span_R}(p, q)$ spanned by the real and imaginary parts of the initial state). Since $\delta = 10^{-15}$, Π is essentially **the plane spanned by the parent v** .

(Eq. 23) **Measured quantities** — For the state z at each step:

$$H_{\perp}/H = \| z - p(p \cdot z) - q(q \cdot z) \|^2 / \| z \|^2 \quad (\text{energy fraction outside the readout plane } \Pi)$$

$$H_{\text{total}} = z^\dagger z$$

$$\text{closure} = |z^\dagger T z| / z^\dagger z$$

(Eq. 23') **Why $H_{\perp}/H$ is the “dormant fraction” and does not pick up the parent’s motion**

— The orbit of the pure parent ($\delta=0$) is a relative equilibrium (Eqs. 15', 19): $Z(t) = e^{\{-i\sigma t\}} Z(0)$. Expanding,

$$e^{\{-i\sigma t\}}(x + iy) = (x \cos \sigma t + y \sin \sigma t) + i(y \cos \sigma t - x \sin \sigma t), \quad x = \text{Re } Z(0), \quad y = \text{Im } Z(0)$$

i.e., the real and imaginary parts always remain within $\text{span_R}(x, y) = \Pi$. The complementary orthogonal projection $z - p(p \cdot z) - q(q \cdot z)$ exactly annihilates any complex combination inside Π , so **$H_{\perp}$ is identically 0 along the parent orbit**. Hence $H_{\perp}/H$ measures only “everything **other than** the parent’s rigid rotation” — initially the out-of- Π component of the seed g ($H_{\perp}/H(0) \approx \delta^2 \sim 10^{-30}$), and afterwards whatever the dynamics grows from it. Because the seed is energy invisible at the parent’s scale, the July canonical calls this the **dormant fraction**.

(Eq. 23'') **Identity of definition with the July canonical** — The $H_{\perp}/H$ of this chapter is the same measurement as $f(\tau)$ of the July canonical. Canonical program 自発の分裂予備実験_v1/run_spontaneous_splitting_largeN_v1.py:

```

57         Zp = Z - p * (p @ Z) - q * (q @ Z)
58         htot = float(np.real(np.conj(Z) @ Z))
59         f = float(np.real(np.conj(Zp) @ Zp)) / htot

```

(The construction of p, q is also identical; lines 46–48.) The curves of this chapter therefore plot the same quantity with the same definition as the July inflation figure (dormant_growth_large_n_v1.png) and can be compared directly.

Justification as the inflation indicator — By the above, the time evolution of $H_{\perp}/H$ quantifies inflation-like development by three points: “(i) the initial value lies at the seed scale 10^{-30} (as long as the parent is an equilibrium, there is no jump at the first step); (ii) from there a straight line on the semilog plot — i.e., constant-rate exponential amplification — continues; (iii) it saturates at $O(10^{-2-1})$.” When (i) fails (a jump to $10^{-8-10-3}$ at the first step), that is a mismatch injection and is distinguished from seed amplification (the deletion controls of Appendix A are discriminated precisely by this (i)). The onset (Eq. 24) is the indicator of reaching (iii).

Conservation: by Eq. 19 the map is real orthogonal, so both the numerator $|z^\dagger T z|$ and denominator $z^\dagger z$ of closure are theoretically invariant; the time variation of closure measures only the accumulation of

numerical rounding (measured in §6, G3). Norm conservation guarantees $H \perp / H \in [0,1]$.

(Eq. 24) onset (bookkeeping indicator) — The first step number at which $H \perp / H > 0.05$ (– 1 if none). A tally that does not affect the dynamics.

2.6 Correspondence with the Complex-Plane Readout — Design Intent of the Inflation

Figure and the Three Complex-Plane Figures

The complex-plane figures of Chapter 3 (step 0, final step, zoom into the condensed center) are the readout paired with the inflation figure ($H \perp / H$ curves) of this chapter. The identity that fixes their relation is given here first.

Caution (two easily confused “planes”) — The complex plane of the figures is “the plane on which the M per-edge values $z_e \in \mathbb{C}$ are plotted”, whereas the parent plane $\Pi = \text{span}_R(p,q)$, the reference of $H \perp / H$, is “a real 2-dimensional subspace of the state space $\mathbb{C}^M$ ”. They are distinct, but connected by the following identity.

(Eq. 25) Per-edge representation of states inside Π — With $x = \text{Re } z_0$, $y = \text{Im } z_0$, $v \approx z_0$ ($\delta = 10^{-15}$), $x = (v + \bar{v})/2$ and $y = (v - \bar{v})/(2i)$ give

$$z \in \text{span}_{\mathbb{C}}\{x, y\} = \Pi \iff z_e = a \cdot v_e + b \cdot \text{conj}(v_e) \quad (a, b \in \mathbb{C} \text{ constants independent of the edge})$$

That is, ****motion confined to Π can, in the figure, produce only deformations within the two-complex-parameter family “uniform rotation/scaling of the parent’s star ($a \cdot v_e$) plus admixture of its mirror image ($b \cdot \bar{v}_e$)”.** In particular the pure parent orbit $Z(t) = e^{\{-i\sigma t\}} Z(0)$ (Eq. 15’) has $a = e^{\{-i\sigma t\}}$, $b = 0$: the whole star rotates uniformly about the origin without changing its shape**. Conversely, the appearance in the figure of shapes not expressible in this family (e.g., phase dispersal in all directions = a ring) is the configurational expression of the state having left Π .

By this identity, the three complex-plane figures share the roles of corroborating, from the configuration side, the start point, end point, and interior of the end point of the inflation figure ($H \perp / H$ curves):

Figure	Correspondence to the $H \perp / H$ curve	What it reads
step-0 figure	start of the curve ($H \perp / H \approx 10^{-31}$)	the contents of Π = the parent’s shape (two antipodal pairs, 4-bundle star). That the step-1 figure is indistinguishable from step 0 visualizes $f(1)$ staying at seed scale (§2.5 (i))
final-step figure	saturation region of the curve ($H \perp / H \sim 0.05\text{--}0.46$)	departure to shapes not expressible in the family of Eq. 25 (star $\rightarrow$ ring) = configuration in which out-of- Π components dominate. onset (Eq. 24) roughly corresponds to when the shape breakdown becomes visible

Figure	Correspondence to the $H \perp/H$ curve	What it reads
zoom into the condensed center	complement of what the curve does not say	$H \perp/H$ measures only “how much left Π ”. The zoom resolves “the organization at the destination” — whether angular clusters are condensation onto exactly identical complex values (exact degeneracy) or bundles of finite width

The implementation of the figures (grid rendering, cluster-extraction algorithm, line-number correspondence) is described in Chapter 3.

3. Implementation Method

- The sweep body `run_N3_N40_stage123_v1.py` is self-contained (no engine import). The bundled engine `run_n_scaling_lowrank_v1.py` serves only as the definition body of §2.2 (the old mathematics) and for the parent generation of Chapter 1; it plays no role in this chapter’s dynamics loop.
- This program is derived from the ChatGPT-authored canonical `run_and_plot_N3_N40_mixedseed_20260903.py` through **a documented chain of minimal diffs only** (Appendix B); the dynamics function `one_step` is identical to the stage-3 version that passed the three gates in the $N=40$ single-factor experiments.
- The initial data are the Z0 of the Chapter-1 static-parent npz files, loaded per N (no regeneration, no modification).
- The input gate `check_sweep_inputs_v1.py` verifies post hoc that the stored $Z[0]$ of all 228 runs is bit-identical to the static parents’ Z0.
- The numbers in §6 are transcribed from the output of the aggregation program `analyze_sweep_summary_v1.py`, `results/analysis_sweep_summary_v1.json` (no manual computation or manual tallying).

Stage $\Leftrightarrow$ equation $\Leftrightarrow$ implementation correspondence table (implementation: `run_N3_N40_stage123_v1.py`):

Stage	Content	Equations	Implementation
Stage 1	explicit matrix + exact spectral map	Eq. 16	lines 21–22 (<code>H_of</code>), lines 27–28 (<code>eigh</code> and application of the phase factor)
Stage 1	fixed $\Delta \tau = 2 \pi / \text{den}$ clock and denominator series	Eq. 16	<code>2.0*math.pi/den</code> on line 27; line 38 (series generation)
Stage 2	amplitude normalization ($z = e^{i \theta}$)	Eq. 17	<code>np.exp(1j*np.angle(z))</code> on line 26

Stage	Content	Equations	Implementation
Stage 3	imaginary-part extraction $H_3 = i \cdot \text{Im}(\hat{H}) \rightarrow \text{real}$ orthogonal rotation	Eqs. 18, 19	<code>(1j*np.imag(H))</code> on line 26
—	readout plane, measured quantities, onset	Eqs. 22, 23, 24	lines 29–30, 31–32, 45–46

(Contrast: the old Eqs. 12–14 exist only in engine lines 122–132 and 134–141 and are never called in this chapter’s loop.)

4. Detailed Design

4.1 Overall Flow

for N in 3..40:

 [initialization] load static parent Z0 $\rightarrow$ adjacency A (Eq. 10) $\rightarrow$ readout plane p,q (Eq. 22)
 $\rightarrow$ denominator series (Eq. 16)

 for den in series:

 [loop] t=0..500: record state and measured quantities (Eq. 23); apply one_step (Eqs. 16–19) if t<500

 [finalization (den)] save state npz; append onset (Eq. 24) etc. to the summary row
 [finalization (all)] write timeseries/summary CSVs $\rightarrow$ 8×5 grid figure $\rightarrow$ RUN_METADATA

4.2 Overall Data Flow

- **Input:** Z0 from `parents/parent_static_N{N:05d}_makeparent_20260905.npz` (38 files; Chapter-1 products; SHA256 canonical in SHA256SUMS.txt)
- **Parameters:**
 - STEPS = 500 ... steps per run (no early stopping; fixed)
 - OFFSETS = (-2,-1,0,1,2) ... offsets of the denominator series from N (Eq. 16)
 - denominator 124 ... the control denominator common to all N
 - dtype: state complex128 / real float64 (fixed by the assert on line 12)
- **Output:**
 - `results/hm_N{N}_den_{den}_states_500.npz` $\times$ 228 (Z(501×M), N, denominator, steps)
 - `results/timeseries_64bit_with124_N3_N40.csv` (columns: N, series, denominator, step, Hperp_frac, H_total, global_closure; 228×501 = 114,228 rows)
 - `results/summary_64bit_with124_N3_N40.csv` (columns: N, series, denominator, onset_gt_0.05, initial, step1, final, max; 228 rows)
 - `results/fig_Hperp_denominator_controls_with_124_N3_N40_stage123.png` (target figure)
 - `results/RUN_METADATA_N3_N40_stage123.json`

4.3 Individual Processes

■4.3.1 Initialization (run_N3_N40_stage123_v1.py) Constants and dtype fixing:

```
11 STEPS=500; OFFSETS=(-2,-1,0,1,2)
12 assert np.dtype(np.float64).itemsize==8 and np.dtype(np.complex128).itemsize==16
```

Edges and adjacency (Eqs. 1, 10):

```
14 def edges(N):
15     a,b=np.triu_indices(N,k=1); return a.astype(np.int64),b.astype(np.int64)
16 def adjacency(N):
17     ea,eb=edges(N); M=len(ea); A=np.zeros((M,M),dtype=np.float64)
18     for e in range(M):
19         share=(ea==ea[e])|(ea==eb[e])|(eb==ea[e])|(eb==eb[e]); share[e]=False; A[e,share]=1.0
20     return A
```

Parent loading, readout plane (Eq. 22), denominator series (Eq. 16):

```
34 for N in range(3,41):
35     # 初期データ: 各 N の静的親ファイルの Z0 を使用
36     z0=np.array(np.load(os.path.join(PARENT_DIR,f'parent_static_N{N:05d}_makeparent_20260905
37     A=adjacency(N); p,q=plane(z0)
38     pairs=[(N+o, f'N{o:+d}' if o else 'N') for o in OFFSETS if N+o>0] + [(124,'124')]
29 def plane(v):
30     p=v.real.astype(np.float64,copy=True); p/=np.linalg.norm(p); q=v.imag.astype(np.float64,
```

■4.3.2 Loop (the one-step dynamics — Eqs. 16–20)

```
21 def H_of(z,A):
22     H=A*(np.conj(z)[: ,None]*z[None, :]); np.fill_diagonal(H,0.0); return H.astype(np.complex128)
23 def one_step(z,A,den):
24     # 段 3 の最小変更 (唯一の力学変更点): 位相のみ生成子  $\hat{H}$  の虚部だけを取る  $H=i \cdot K$  ( $K=\sin(\Delta \theta)$ ) 実反対称
25     #  $\exp(-i \Delta \tau \cdot iK)=\exp(\Delta \tau \cdot K)$  の実直交回転となり、 $Z \sim T Z$  (零閉塞) と  $\|Z\|$  を厳密保存する。
26     H=H_of(np.exp(1j*np.angle(z)),A); H=(1j*np.imag(H)).astype(np.complex128,copy=False)
27     w,V=np.linalg.eigh(H); phase=np.exp(-1j*np.float64(2.0*math.pi/den)*w)
28     return (V@(phase*(V.conj().T@z))).astype(np.complex128,copy=False)
```

- First half of line 26, `np.exp(1j*np.angle(z))`: **stage 2** = Eq. 17 (amplitude normalization; the generator input is replaced by $z=e^{i\theta}$).
- Second half of line 26, `(1j*np.imag(H))`: **stage 3** = the imaginary-part extraction of Eq. 18 $\rightarrow$ the generator $H_3=i \cdot K$ of Eq. 19 (`np.imag` is the elementwise imaginary part; the imaginary part of the Hermitian $\hat{H}$ is the real antisymmetric K).
- Lines 27–28: **stage 1** = Eq. 16 (exact spectral decomposition by `np.linalg.eigh`; application of the phase factor $e^{-i(2\pi/\text{den})w}$ with fixed $\Delta \tau=2\pi/\text{den}$). The eigenplane rotation angle is Eq. 20, $\psi_k=(2\pi/\text{den})\sigma_k$, which differs from the old Eq. 14 (arctan compression, $\sigma_{\max}$

normalization) (§2.4).

Measurement and recording (Eq. 23):

```

31 def metrics(z,p,q):
32     h=np.vdot(z,z).real; zp=z-p*np.dot(p,z)-q*np.dot(q,z); hp=np.vdot(zp,zp).real; return fl

40     z=z0.copy(); vals=np.empty(STEPS+1,np.float64); states=np.empty((STEPS+1,z.size),np.
41     for t in range(STEPS+1):
42         states[t]=z; vals[t],htot[t],closures[t]=metrics(z,p,q)
43         if t<STEPS: z=one_step(z,A,den)

```

Stopping conditions and exceptional behavior (elements not formulated as equations): -

There is no early stopping. Every run consists of a **fixed 501** recordings ($t=0..500$) and 500 applications of the map (lines 41–43). onset (Eq. 24) is used only for tallying and never stops the dynamics (the `np.flatnonzero(vals>0.05)` on lines 45–46). - For runs at larger N , numpy RuntimeWarnings (divide by zero / overflow / invalid value encountered in matmul) may be printed for the matrix product $V @ (\text{phase} * (V^\dagger z))$. These are warnings, not exceptions; execution does not stop. In this series, bit-identity of reruns on identical inputs (Chapter 1 G1 and this chapter’s input gate) confirms they do not affect the results. - For $N=3,4$ the $N+o>0$ filter of Eq. 16 admits small denominators such as $\text{den}=1$ ($\Delta \tau = 2 \pi$) into the series. There is no special-casing in the program (a consequence of the comprehension on line 38). From the viewpoint of Eq. 20, small den means large $\Delta \tau$ — the regime where aliasing ($\Delta \tau \sigma_k \bmod 2 \pi$) acts strongly.

■4.3.3 Finalization (saving, plotting, metadata)

```

44     np.savez_compressed(os.path.join(OUT,f'hm_N{N}_den_{den}_states_500.npz'),Z=states,N
45     rows.extend((N,label,den,t,vals[t],htot[t],closures[t]) for t in range(STEPS+1)); ix
46     summaries.append((N,label,den,int(ix[0]) if ix.size else -1,float(vals[0]),float(val

```

The CSVs, figure, and metadata (lines 48–66) only write out measured values and render the figure; they do not affect the dynamics. The figure is an 8×5 grid (lines 55–64), 38 panels used and 2 panels off (line 64).

■4.3.4 Input Gate (check_sweep_inputs_v1.py)

```

18 for N in range(3, 41):
19     Z0 = np.load(os.path.join(PARENT_DIR, f'parent_static_N{N:05d}_makeparent_20260905.npz'))
20     dens = [N + o for o in (-2, -1, 0, 1, 2) if N + o > 0] + [124]
21     for den in dens:
22         p = os.path.join(RESULT_DIR, f'hm_N{N}_den_{den}_states_500.npz')
23         same = bool(np.array_equal(np.load(p)['Z'][0], Z0))
24         n_checked += 1
25         if not same:
26             print(f'MISMATCH: N={N} den={den}')
27         ok = False
...

```

```
35 sys.exit(0 if ok else 1)
```

It verifies that the $Z[0]$ of every stored npz is `np.array_equal` (bit-identical) to the corresponding static parent Z_0 ; a single mismatch yields exit code 1.

5. Execution Results

5.1 Reproduction Commands

```
cd N3_N40_stage123_sweep_20260905
python3 run_N3_N40_stage123_v1.py      # sweep body
python3 check_sweep_inputs_v1.py      # input gate
python3 analyze_sweep_summary_v1.py    # aggregation for the numbers in §6
# or ./run_all.sh (parents → sweep → gate → aggregation → plots)
```

5.2 Execution Environment

- Python 3.9.6 (`.venv/bin/python3`), numpy 2.0.2 (BLAS/LAPACK: macOS Accelerate), matplotlib (grid rendering)
- macOS 26.3.1 (arm64)

5.3 Execution Time

About **45–50 minutes** for the sweep body ($38\text{ N} \times 6\text{ denominators} \times 500\text{ steps}$, including full state saving; measured 2026-09-05; dominated by the $M \times M$ `eigh` $\times 500 \times 6$; $N=40$ alone about 3 minutes). Input gate and aggregation: under 1 minute each.

5.4 Verification Gates

Gate	Pass condition	Measured	Verdict
G1: input identity	$Z[0]$ of all 228 npz bit-identical to the corresponding static parent Z_0	checked 228 / MISMATCH 0	PASS
G2: completion	ALL DONE printed, exit code 0	confirmed	PASS
G3: conserved quantity (observation)	$\text{closure} = z^{\text{T}}z /H$ stays small over all runs and steps	step-0 max 5.19e-15; global max 3.72e-13	PASS

(The G3 numbers are `global_closure_step0_max` / `global_closure_all_max` of `analysis_sweep_summary_v1.json`.)

5.5 Data

Item	Content
Folder	N3_N40_stage123_sweep_20260905/results/
State npz	hm_N{N}_den_{den}_states_500.npz $\times$ 228 (tens of KB to $\sim$ 5.9MB each; actual sizes per SHA256SUMS.txt and the bundle)
Timeseries CSV	timeseries_64bit_with124_N3_N40.csv ($\sim$ 8.7MB, 114,228 rows)
Summary CSV	summary_64bit_with124_N3_N40.csv (228 rows)
Metadata	RUN_METADATA_N3_N40_stage123.json
Aggregation	analysis_sweep_summary_v1.json (source of the numbers in §6)
SHA256	Canonical values for all files in the bundled SHA256SUMS.txt. Sweep program: 1abf2353fee2e4f56f05e7a6f149fd086885136beb61ab571b48a56b run_N3_N40_stage123_v1.py

5.6 Figures

- Target figure: results/fig_Hperp_denominator_controls_with_124_N3_N40_stage123.png (8 $\times$ 5 grid; the **vertical axis of each panel is the dormant fraction $H \perp / H$** [Eq. 23; meaning in §2.5] on a semilog scale, horizontal axis is the step; the six denominator curves are overlaid. Straight segments on the semilog correspond to inflation-like exponential amplification; horizontal segments to saturation.)
- The complex-plane readout figures (step 0, final, zoom) are treated in Chapter 3.

6. Execution Analysis (objective report and observations only; numbers from

analysis_sweep_summary_v1.json)

1. Of the 228 runs, **212 crossed $H \perp / H > 0.05$ within 500 steps** (onset range 45–481).
2. **In all 228 runs, step 1 remained at seed scale:** $H \perp / H(0) \in [1.92\text{e-}33, 1.60\text{e-}31]$, $H \perp / H(1) \in [3.64\text{e-}30, 7.32\text{e-}25]$. There were **zero** mismatch injections (jumps of order $10^{-8-10-3}$).
3. Final values of the crossing runs: 0.0501–0.458.
4. Breakdown of the 16 non-crossing runs: 15 runs saturated with final 0.025–0.048, **just below the 0.05 threshold** (the 124 series and N+2 series for $N \geq 26$, and the three series of $N=39$); only one run was actually slow-growing: $N=3$, $\text{den}=124$ (final $1.75\text{e-}8$).
5. The onset of the 124 series shortens from 318 at $N=4$ as N increases, entering the band 89–160 for $N \geq 15$ (crossing N only; the table is `onset_by_series['124']` of `analysis_sweep_summary_v1.json`).
6. The zero closure $|z^{\wedge}\text{Tz}|/H$ stayed at most $3.72\text{e-}13$ over all 228×501 recorded points (initial max $5.19\text{e-}15$) — a numerical confirmation of the conservation law of Eq. 19.
7. Visual observation of the figure: every panel shows the same family of curves — “straight exponential amplification from seed scale $\rightarrow$ saturation in the $10^{-3-10-1}$ range”.

Appendix A: Provenance of the Stage Composition — Audit Table of the Single-Factor

Experiments and Deletion Controls

The dynamics of this chapter (stages 1+2+3) is the composition established by the single-factor experiment series on N=40 with the static parent (the same Z0 as Chapter 1). The complete set of experiments (programs, data, figures, README, SHA256SUMS) is contained in `ChatGPT_denominator_controls_N40_selfcontrol_20260904/` and is **bundled with the upload of this paper**. The numbers originate from the summary CSVs and READMEs of the respective results directories.

Composition	Dynamics (generator / clock)	Level of f(1)	0.05 crossing (500 steps, 6 dens)	Final values / observation	Output directory
Stage 1 only (baseline)	$H=A^*(\bar{z} \otimes z)$ (amplitude- weighted, cos included) / fixed $\Delta \tau$	5.3–6.5e-8 (injection)	0/6	stable at ceiling $\sim 1.16\text{e-}3$	results_static- parent/
Stages 1+2 (stage 3 deleted)	phase-only $\hat{H}$ (cos included) / fixed $\Delta \tau$	1.4e-9–8.9e-3 (injection, den- dependent)	6/6 (τ =4–198)	departure up to 0.94–0.9999	results_static- par- ent_phaseonly/
Stages 1+3 (stage 2 deleted)	amplitude- weighted $i \cdot \text{Im}(H)$ / fixed $\Delta \tau$	1.8–2.3e-8 (injection)	0/6	slow drift to $\sim 3.5\text{e-}3$	results_static- par- ent_ampimK/
Stages 1+2+3 (this chapter)	phase-only $i \cdot K$ / fixed $\Delta \tau$	4.0–9.1e-29 (seed scale)	4/6 (τ =276–456)	0.038–0.10, relaxation curve present	results_static- parent_imK/
Stage 2 at init only (reference)	z0 equimodu- larized once + amplitude- weighted $i \cdot \text{Im}(H)$ / fixed $\Delta \tau$	2.9–3.4e-3 (injection)	6/6 (τ =3–6)	0.96–0.99, immediate departure	results_static- par- ent_stage2init/
Stages 2+3 + σ clock (reference)	phase-only $i \cdot K$ / σ -normalized Cayley clock	5.84e-30 (seed scale)	0/6 (within 500 steps)	late-segment slope 65.8 steps/decade (July canonical: 64.0)	results_static- parent_sig- maclock/

Observation (facts only): f(1) remains at seed scale ($\sim 10^{-29}$) **only** in the composition where stages 2 and 3 are present **simultaneously**; deleting either one makes f(1) jump to $10^{-8-10-3}$. Moving stage 2 to a

one-time initialization produces the largest jump (10^{-3}).

Appendix B: Program Lineage (chain of minimal diffs and SHA256)

The diffs of each step are recorded in full in the READMEs of the respective packages.

#	Program	Diff from the previous
0	ChatGPT_denominator_controls_N3_N40_mixedseed_20260903/run_and_plot_N3_N40_only_selfcontrol_20260904 (ChatGPT-authored canonical)	
1	ChatGPT_denominator_controls_N40_only_selfcontrol_20260904/run_and_plot_N40_only_selfcontrol_20260904	range(40,41) (N=40 output verified bit-identical to canonical)
2	... /run_N40_staticparent_v1.py	initial data replaced by the Chapter-1 static parent Z0 (input only)
3	... /run_N40_staticparent_phaseonly_v1.py	1 line in one_step: H built from only $\text{pyg } z$ (stage 2)
4	... /run_N40_staticparent_imK_v1.py	1 line in one_step: $H=1j \cdot \text{Im}(\hat{H})$ (stage 3)
5	N3_N40_stage123_sweep_20260905/run_and_plot_N3_N40_stage123_v1.py (this chapter)	loading, output destination, figure/metadata names only (dynamics unchanged)

SHA256 (full digits; transcribed from the output of shasum -a 256):

c709c56335d4c67373bff9a3ef6414ea17564d5fbf9c10b7bc9c3724ff091b92	run_and_plot_N3_N40_mixedseed_20260903
5a07f354e19985dca0f5de89217e2aa22ac511afaa4b2bb4aa5d93e0c7f9706f	run_and_plot_N40_only_selfcontrol_20260904
cb5a0ab6db9ae5719eac7aee3b539eb04ca09f5ebfcab6dcce36e8a6e727719e	run_N40_staticparent_v1.py
c1d6d2e60e101f0a99585eefdc22990a087c46246aba112042ac97a7fcbda71	run_N40_staticparent_phaseonly_v1.py
a67912b77f7f112731c1eac7612f21b464aed7e2c42431f7b3a7afabb2cd051d	run_N40_staticparent_imK_v1.py
1abf2353fee2e4f56f05e7a6f149fd086885136beb61ab571b48a56b09691567	run_N3_N40_stage123_v1.py

(End of Chapter 2. Chapter 3, “Complex-Plane Readout Figures”, follows.)